

LAHIRUNI MALSHIKA

Trainee Software Engineer | Final Year IT Undergraduate

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ABOUT ME

Trainee Software Engineer and Final-Year IT undergraduate, bringing hands-on experience across full-stack web development, mobile applications, and AI-driven solutions. My journey through industry experience, academic projects, and self-driven builds has sharpened my skills in MERN stack, Flutter, and DevOps, with a focus on clean code architectures and real-world problem-solving. Seeking a Software Engineer role to contribute effectively to innovative projects while continuously growing as a software engineering professional.

WORK EXPERIENCE

Trainee Software Engineer at Virstack

DEC 2025 - JUN 2026

- Contributed to developing and maintaining AI-powered full-stack applications using the MERN stack, alongside cross-platform mobile development with Flutter.
- Responsible for optimizing application performance to ensure fast, reliable, and efficient user experiences across web and mobile platforms and Integrated AI-driven features to enhance product functionality and built scalable interfaces with clean code architectures.
- Supported DevOps workflows through Docker containerization and AWS cloud infrastructure, collaborating within an agile team using version control to deliver production-ready solutions.

Intern Software Engineer at Virstack

MAR 2025 - NOV 2025

- Contributed to full-stack web and mobile application development using React, Angular, Express with TypeScript, and Flutter (Dart). Built responsive UI components, implemented form validation and API integrations, and supported feature development in a production-level environment.

EDUCATION

University of Moratuwa

Bsc.(Hons) in Information Technology

JUN 2022 - JUL 2026

- Dean's List - Level 2 Semester 1 (SGPA: 3.89)
- Dean's List - Level 2 Semester 2 (SGPA: 3.94)
- Current GPA - 3.43 (out of 4.0 - up to L3S2)

Sujatha Vidyalaya Matara

G.C.E. Advanced Level

OCT 2020

- Stream - Physical Science

BritishWay English Academy

Diploma in English

2021

TECHNICAL SKILLS

Programming Languages : Java, C, Python, JavaScript, TypeScript, Dart

Frontend Development : React.js, Next.js, JavaScript, Tailwind CSS, HTML, CSS, Flutter

Backend Development : Node.js, Express.js, Nest.js

Databases : MySQL, MongoDB, Neo4j

Cloud & Deployment : Docker, Vercel, CI/CD (GitHub Actions), AWS Lightsail, AWS S3, AWS EC2

Testing & Quality Assurance : Cypress, Selenium, Unit Testing, End-to-End Testing

Tools : Git, Postman, SonarQube

PROJECTS

Final Year Research - Exploring Hardware-Software Codesign of Spatial Deep Learning Hardware

Accelerators (Group project - Ongoing)

Research on developing a cycle-accurate hardware-software co-design simulator framework for systolic array-based DNN accelerators, addressing critical performance bottlenecks in DNN execution on computing systems. Implementing SystemVerilog RTL models coupled with a Python compiler/mapper, memory management schemes, exploring loop optimizations, and multi-DNN scheduling algorithms.

- Technologies - Python, Verilog, DNN Workloads, FPGA Simulation

Fuel Price Prediction Dashboard using Machine Learning (Individual Project)

Built a machine learning model to predict commonly used fuel prices in Sri Lanka. Performed data preprocessing, feature engineering, and classification.

- Technologies: Python, Machine Learning, Pandas, Scikit-learn
- FuelPricePrediction Dashboard Repository

Travel Recommendation Expert System (Individual project)

Developed a rule-based expert system to recommend travel destinations using logical inference. Built a Prolog knowledge base for destinations, costs, seasons, and ratings. Integrated the Prolog reasoning engine with a Python GUI. Implemented features like best season prediction and cheapest destination finder.

- Technologies - Python, Prolog (SWI-Prolog), Tkinter
- ExpertSystem-TravelAdvisor Repository

MozzaMelt - Ecommerce Platform (Group Project)

A web application for a pizza shop, enabling users to order pizzas, make payments, track orders, manage profiles, and provide reviews. I have contributed to the front-end development and testing.

- Role - Frontend Developer, Quality Assurance Engineer
- Technologies - React Js, Asp.Net Core, MsSql, Docker, Ci/Cd Pipeline, Selenium
- Frontend Repository • Backend Repository

Anothershots - Photography Focused Web Application (Level 2 Software Project - Mentored by 99X)

An inclusive platform that helps photography professionals showcase their skills and connect with potential clients, benefiting both photographers and users seeking high-quality photography services. I have implemented the Featured Photo Section, Contact Section, and the Packages Section in the Photographer's profile, Photographer's Booking, and Admin Dashboard.

- Role - Full Stack Developer
- Technologies -Next JS, TypeScript, Tailwind CSS, Nest JS, Prisma ORM, MongoDB
- Anothershots Frontend Repository • Anothershots Backend Repository

Image Search Application (Individual Project)

I developed a streamlined image-search application with a user-friendly interface that enables users to search for images by entering relevant keywords.

- Role - Full Stack Developer • Image Search App Repository
- Technologies - React JS, Tailwind CSS, Unsplash API

ACHIEVEMENTS AND PARTICIPATIONS

- **CyberZee'24 - Top 06** : An inter university quiz competition organized by University of Kelaniya.
- **ACES Coders V11.0 2024 - Participant** : An inter university coding competition organized by University of Peradeniya.
- **MoraXtreme8.0 2023 - Top 75** : An inter university coding competition organized by IEEE student branch of University of Moratuwa.
- **Code Rush 2023 - Intra Faculty Coding Competition** : Participation
- **RealHack 5.0 - Participant** : An inter university coding competition organized by University of Kelaniya.

EXTRA CARICULAR ACTIVITIES

- **IEEE Student Branch of UoM** : Publicity committee member - Rise Up Mora 2023
- **IEEE Power & Energy Society Chapter of UoM** : WebMaster (Term of 24/25)
- **IEEE Women In Engineering Affinity Group of UoM** : Chief Organizer (Term 24/25), Marketing Committee Lead
- **Mathematics Society of UoM** : Publicity committee member - ENIGMA 2024
- **Leo Club of UoM**

REFERENCES

Dr. T.M. Thanthriwatta

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